

midterm 2

Tara Hendrickson

unit 4:

1. a. into the page
b. left
c. out of the page
2. a. closed = counterclockwise, closed long time = no change, open = clockwise
b. closed = clockwise, left closed = no change, open = counterclockwise
c. closed = counterclockwise, left open = no change, open = clockwise

$$3. a. A = \pi (d/2)^2 = \pi (.02/2)^2 = 3.801 \times 10^{-4} \text{ m}^2$$

$$\Delta \Phi_B = B \cdot A = 2 \times 3.801 \times 10^{-4} = 7.602 \times 10^{-4} \text{ Wb}$$

$$EMF_{avg} = 7.602 \times 10^{-4} / 0.250 = \boxed{3.04 \text{ mV}}$$

$$b. I = EMF/R = 3.04 \times 10^{-3} / .01 = \boxed{0.304 \text{ A}}$$

$$c. P = I^2 R = (.304)^2 \cdot .01 = \boxed{9.24 \times 10^{-4} \text{ W}}$$

$$4. a. N = 18 / (.64 \times 3 \times 10^{-4} = 1875)$$

$$= \boxed{267}$$

$$b. T = 2\pi / 1875 = \boxed{3.35 \times 10^{-5} \text{ s}}$$

$$5. a. \frac{N_p}{N_s} = \frac{240}{120} = \boxed{2.00}$$

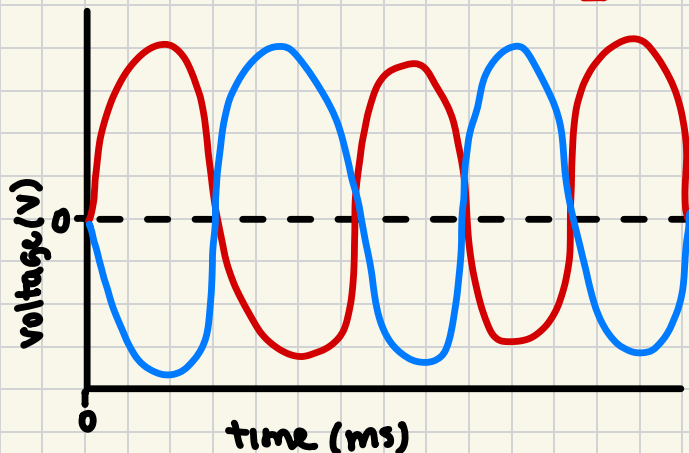
$$b. \frac{I_p}{I_s} = \frac{1}{2} = \boxed{0.50}$$

$$6. a. V(t) = 120 \sin(240\pi t)$$

$$b. = -120 \sin(240\pi t)$$

$$c. t = \frac{n\pi}{240\pi} = \frac{n}{240}$$

$$= \frac{1}{240} \text{ s} = 4.17 \text{ ms}$$



$$7. \Delta t = \frac{L \Delta I}{\mathcal{E}} = \frac{(2 \times 10^{-3}) \cdot .1}{500} = \boxed{4.00 \times 10^{-7} \text{ s}}$$

$$8. a. \mathcal{E} = 25 \cdot (100 / .08) = \boxed{31250 \text{ V}}$$

$$b. \mathcal{E} = \frac{1}{2} \cdot 25 \cdot 100^2 = \boxed{125000 \text{ J}}$$

$$c. P = 125000 / .08 = \boxed{1.56 \times 10^6 \text{ W}}$$

$$d. N^2 = \frac{25 \cdot .5}{(4\pi \cdot 10^{-7}) \cdot .01} = 9.95 \times 10^8 = N = \boxed{31500 \text{ turns}}$$

$$e. X_L = 2\pi f L = 2\pi \cdot 10^4 \cdot 25 = \boxed{1.57 \times 10^6 \Omega}$$

$$9. a. L = (20 \times 10^{-9}) \cdot (5 \times 10^6) = \boxed{0.1 \text{ H}}$$

$$b. R = .1 / (1 \times 10^{-9}) = \boxed{1.00 \times 10^8 \Omega}$$

$$c. \frac{I(t)}{I_0} = e^{-t/\tau} = e^{-.8} = .449 = \boxed{44.9\%}$$

$$10. a. V_{in} - iR = 0 \rightarrow \boxed{V_{in} = iR}$$

$$b. V_{in} = RC (dv_{out} / dt) + V_{out}$$

$$c. \frac{V_{out}}{V_{in}} = \frac{1}{1 + j\omega RC}$$

$$d. \frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + (200\pi)^2}} = \boxed{0.00159}$$

11. a. $f_0 = 1 / (2\pi \sqrt{.01 \times 1 \times 10^{-6}}) = \boxed{1591.55 \text{ Hz}}$

b. at $f = 0.1$: $Z = \sqrt{100^2 + (10 - 1000)^2} = \boxed{995 \Omega}$
 at $f = 10$: $Z = \sqrt{100^2 + (1000 - 10)^2} = \boxed{995 \Omega}$

12. a. $I_{\text{rms}} = 120 / 995 = \boxed{0.121 \text{ A}}$

b. $P_{\text{rms}} = 120 \cdot .121 \cdot .1 = \boxed{1.45 \text{ W}}$

13. carrier = 1.4 MHz, upper sideband = 1.41 MHz, lower sideband = 1.39 MHz

unit 5:

1. a. $B = 4\pi \times 10^{-8} / (2\pi \times .01) = 2 \times 10^{-6} \text{ T} = \boxed{2 \mu\text{T}}$

b. $\boxed{0.1 \text{ A}}$

2. a. $d = (3 \times 10^8) \cdot (5 \times 10^{-6}) = \boxed{1500 \text{ m}}$

b. $R = ((1.78 - 1) / (1.78 + 1))^2 = 0.0787 = \boxed{7.87\%}$

3. a. $I = 1000 \text{ W} / .001 \text{ m}^2 = \boxed{1000000 \text{ W/m}^2}$

b. $I_{\text{av}} = 1000000 \cdot (\frac{1}{4}) = \boxed{250000 \text{ W/m}^2}$

4. a. $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$
 $n_2 \sin(\theta_2) = n_3 \sin(\theta_3)$
 $n_1 \sin(\theta_1) = n_3 \sin(\theta_3) \rightarrow \boxed{\theta_1 = \theta_3}$

b. $M = 1 + (25/12) = \boxed{3.08}$

5. $N = 3.294 \times 10^{22}$

$$M = (3.294 \times 10^{22}) \cdot (100 \times 10^{-24}) = 6.588 \text{ cm}^3$$

$$I/I_0 = e^{-6.988} = 0.00138 = \boxed{0.138\%}$$

6. a. $\text{dose} = (.25/60) \cdot (1/.01) = \boxed{0.417 \text{ rad}}$

b. $\text{dose} = (.25/2) \cdot (1/.01) = \boxed{12.5 \text{ rad}}$

c. $\boxed{0.125 \text{ Sv}}$

d. not life threatening but could cause biological defects